

IMCI extended & young-infant branches — clinician review

IMCI · clinical sign-off

UNREVIEWED. These classifications and dose bands train a model only with `--include-extended` (off by default). Nothing ships until signed off against the current national IMCI adaptation. Every row here is generated from the code, so it matches exactly what the model was trained to say.

Sources: `data/imci_2022/classifications.json` (2022 SA adaptation, cross-checked vs the WHO 2014 generic); doses in `data/imci_2022/dosing_tables.json`. Severity uses the IMCI colour tiers. To review: confirm each **trigger** and **severity**, tick **OK** or write a correction, then sign at the end.

Classifications — severity, trigger & action (as implemented)

Fever / malaria

Classification	Severity	Trigger (as coded)	Action	Drugs	OK
<code>very_severe_febrile_disease</code>	PINK – refer	Fever with a general danger sign or a stiff neck classifies as very severe febrile disease, whatever the malaria test shows.	Give the first dose of an appropriate antibiotic. Treat the child to prevent low blood sugar. Give paracetamol for high fever. Refer URGENTLY to hospital.	ceftriaxone_ im, paracetamol	
<code>malaria</code>	YELLOW – treat	The malaria test is positive and there are no danger signs or stiff neck.	Give a first-line oral antimalarial as the national guideline directs. Give paracetamol for high fever. Advise the mother when to return immediately. Follow up in 3 days if the fever persists.	artemether_ lumefantrine, paracetamol	
<code>fever_no_malaria</code>	GREEN – home	The malaria test is negative, so this fever is not malaria.	Look for and treat another cause of fever. Give paracetamol for high fever. Advise the mother when to return immediately. Follow up in 3 days if the fever persists.	paracetamol	
<code>fever_malaria_test_required</code>	YELLOW – treat	This child has fever in a malaria risk area, or has travelled to one, and no malaria test has been done. The chart booklet requires a malaria test before the fever can be classified.	Do a malaria test (RDT or microscopy) now and classify on the result. Give paracetamol for high fever meanwhile, and refer urgently if any danger sign appears.	paracetamol	

Corrections for Fever / malaria:

Measles

Classification	Severity	Trigger (as coded)	Action	Drugs	OK
<code>severe_complicated_measles</code>	PINK – refer	A generalised rash with cough, runny nose, or red eyes meets the measles case definition. There is clouding of the cornea, which makes this severe complicated measles.	Give vitamin A. Give the first dose of an appropriate antibiotic. If there is clouding of the cornea or pus draining from the eye, apply tetracycline eye ointment. Refer URGENTLY to hospital.	vitamin_a	

Classification	Severity	Trigger (as coded)	Action	Drugs	OK
measles_with_eye_or_mouth_complications	YELLOW – treat	A generalised rash with cough, runny nose, or red eyes meets the measles case definition. There is pus draining from the eye.	Give vitamin A. If there is pus draining from the eye, apply tetracycline eye ointment. If there are mouth ulcers, treat with gentian violet. Follow up in 3 days.	vitamin_a	
measles	GREEN – home	A generalised rash with cough, runny nose, or red eyes meets the measles case definition. There are no eye or mouth complications and no danger signs.	Give vitamin A. Advise the mother when to return immediately.	vitamin_a	

Corrections for Measles:

Anaemia

Classification	Severity	Trigger (as coded)	Action	Drugs	OK
severe_anaemia	PINK – refer	There is severe palmar pallor.	Refer URGENTLY to hospital.	–	
anaemia	YELLOW – treat	There is some palmar pallor.	Give iron. Give mebendazole if the child is 1 year or older and has not had a dose in the last 6 months. Advise the mother when to return immediately. Follow up in 14 days.	–	
no_anaemia	GREEN – home	There is no palmar pallor.	No treatment for anaemia is needed.	–	

Corrections for Anaemia:

Acute malnutrition

Classification	Severity	Trigger (as coded)	Action	Drugs	OK
complicated_severe_acute_malnutrition	PINK – refer	There is oedema of both feet, which is always severe acute malnutrition. With oedema of both feet, this is complicated severe acute malnutrition.	Give the first dose of an appropriate antibiotic. Treat the child to prevent low blood sugar. Keep the child warm. Refer URGENTLY to hospital.	vitamin_a	
uncomplicated_severe_acute_malnutrition	YELLOW – treat	MUAC is 110mm, below 115mm. The appetite test passed and there is no medical complication.	Give ready-to-use therapeutic food (RUTF) for the child to take at home. Assess and counsel on feeding. Advise the mother when to return immediately. Follow up in 7 days.	amoxicillin, vitamin_a	
moderate_acute_malnutrition	YELLOW – treat	MUAC is 120mm, between 115mm and 125mm.	Assess and counsel on feeding. Advise the mother when to return immediately. Follow up in 30 days.	vitamin_a	

Classification	Severity	Trigger (as coded)	Action	Drugs	OK
no_acute_malnutrition	GREEN – home	There is no oedema of both feet and no wasting by MUAC or weight-for-height.	If the child is less than 2 years old, assess and counsel on feeding. No treatment for acute malnutrition is needed.	–	

Corrections for Acute malnutrition:

Wheeze (cough sub-branch)

Classification	Severity	Trigger (as coded)	Action	Drugs	OK
wheeze_with_danger_sign	PINK – refer	The child has wheeze. There is a general danger sign, so give salbutamol and refer urgently.	Give salbutamol by spacer. Give the first dose of an appropriate antibiotic and refer URGENTLY to hospital.	–	
wheeze	YELLOW – treat	The child has wheeze.	Give salbutamol by spacer for 5 days. Follow up in 5 days if the child is still wheezing. If a cough lasts more than 14 days or the wheeze is recurrent, assess for TB or asthma.	–	

Corrections for Wheeze (cough sub-branch):

Persistent diarrhoea (>=14 days)

Classification	Severity	Trigger (as coded)	Action	Drugs	OK
severe_persistent_diarrhoea	PINK – refer	Diarrhoea has lasted 14 days or more and dehydration is present.	Treat dehydration before referral unless the child has another severe classification. Give an extra dose of vitamin A. Refer URGENTLY to hospital.	vitamin_a	
persistent_diarrhoea	YELLOW – treat	Diarrhoea has lasted 14 days or more with no visible dehydration.	Counsel the caregiver on feeding for persistent diarrhoea. Give an extra dose of vitamin A and give zinc for 14 days. Follow up in 5 days.	zinc, vitamin_a	

Corrections for Persistent diarrhoea (>=14 days):

Dysentery (blood in stool)

Classification	Severity	Trigger (as coded)	Action	Drugs	OK
severe_dysentery	PINK – refer	There is blood in the stool and the child is less than 12 months old.	Treat the child to prevent low blood sugar, keep the child warm, and refer URGENTLY to hospital.	–	
dysentery	YELLOW – treat	There is blood in the stool, the child is 12 months or older, and there is no dehydration.	Treat for 3 days with ciprofloxacin. Advise the mother when to return immediately. Follow up in 2 days.	–	

Corrections for Dysentery (blood in stool):

Sore throat (from 3 years)

Classification	Severity	Trigger (as coded)	Action	Drugs	OK
streptococcal_sore_throat	YELLOW – treat	There is enlarged tonsils, white or yellow exudate on the tonsils, with no runny nose and no cough, which points to a streptococcal sore throat.	Give penicillin. Treat pain and fever. Soothe the throat with a safe remedy. Follow up in 5 days if symptoms are worse or not resolving.	–	
sore_throat_non_streptococcal	GREEN – home	There are not enough signs to classify this as a streptococcal sore throat.	Soothe the throat with a safe remedy.	–	

Corrections for Sore throat (from 3 years):

Growth problem (RTHB curve)

Classification	Severity	Trigger (as coded)	Action	Drugs	OK
growth_problem	YELLOW – treat	On the weight curve, the child is losing weight.	Assess feeding and counsel the caregiver on the feeding recommendations. Deworm and give vitamin A if due. Advise the mother when to return immediately. Follow up in 7 days if there is a feeding problem, otherwise in 14 days.	–	

Corrections for Growth problem (RTHB curve):

HIV

Classification	Severity	Trigger (as coded)	Action	Drugs	OK
confirmed_hiv_infection	YELLOW – treat	The child has a positive HIV test or is already on ART, so HIV infection is confirmed.	Follow the steps to initiate ART. Give cotrimoxazole prophylaxis from 6 weeks of age. Ask about the caregiver's health and manage appropriately. Provide long-term follow-up. ART initiation is not urgent – stabilise any severe illness first.	–	
hiv_exposed	YELLOW – treat	The child is HIV-exposed: on ARV prophylaxis, or with a negative test while still breastfeeding or within 6 weeks of breastfeeding, so infection is not yet ruled out.	Complete the appropriate infant ARV prophylaxis. Repeat HIV PCR testing per the schedule and reclassify on the result. Ask about the caregiver's health and provide follow-up care.	–	
suspected_symptomatic_hiv	YELLOW – treat	There are 3 features of HIV infection (three or more), so symptomatic HIV is suspected.	Counsel and offer HIV testing for the child and reclassify on the result. Counsel the caregiver about her own health and offer testing. Provide long-term follow-up.	–	
possible_hiv_infection	YELLOW – treat	HIV infection is possible because the mother is HIV-positive.	Provide routine care including HIV testing for the child. Counsel the caregiver about her health and offer testing and treatment as needed. Reclassify on the test result.	–	
hiv_infection_unlikely	GREEN – home	The HIV test is negative, all breastfeeding stopped 6 weeks or more before the test, and there are no features of HIV infection.	Provide routine care. Repeat HIV testing only if new features appear or exposure is ongoing.	–	

Corrections for HIV:

Young infant: bacterial infection

Classification	Severity	Trigger (as coded)	Action	Drugs	OK
yi_very_severe_disease	PINK – refer	The young infant has a bulging fontanelle, which is very severe disease.	Give the first dose of an appropriate intramuscular antibiotic. Treat to prevent low blood sugar. Keep the infant warm on the way to hospital. Refer URGENTLY.	–	
yi_local_bacterial_infection	YELLOW – treat	The young infant has a red umbilicus, a local bacterial infection.	Give an appropriate oral antibiotic. If there is eye discharge, give an eye ointment. Teach the caregiver to treat the local infection at home and to give home care. Follow up in 2 days.	–	
yi_no_bacterial_infection	GREEN – home	No signs of very severe disease or a local bacterial infection.	Counsel the caregiver on home care for the young infant.	–	

Corrections for Young infant: bacterial infection:

Young infant: jaundice

Classification	Severity	Trigger (as coded)	Action	Drugs	OK
yi_severe_jaundice	PINK – refer	There is jaundice under 24 hours of age, which is severe jaundice.	Treat to prevent low blood sugar. Keep the infant warm. Refer URGENTLY to hospital.	–	
yi_jaundice	YELLOW – treat	There is jaundice appearing after 24 hours of age, with palms and soles not yellow.	Advise the caregiver to return immediately if the palms and soles become yellow. Follow up in 1 day. If the infant is older than 14 days, refer for assessment.	–	

Corrections for Young infant: jaundice:

Young infant: diarrhoea

Classification	Severity	Trigger (as coded)	Action	Drugs	OK
yi_severe_dehydration	PINK – refer	There is diarrhoea with the infant is less than 1 month old, which is severe dehydration in a young infant.	Start intravenous fluids (Plan C). Give the first dose of an intramuscular antibiotic. Keep the infant warm on the way to hospital. Refer URGENTLY.	–	
yi_dysentery	PINK – refer	There is blood in the stool of a young infant.	Keep the infant warm on the way to hospital. Refer URGENTLY.	–	
yi_severe_persistent_diarrhoea	PINK – refer	Diarrhoea has lasted 14 days or more in a young infant.	Treat any dehydration before referral. Keep the infant warm. Refer URGENTLY.	–	
yi_some_dehydration	YELLOW – treat	There is diarrhoea with two of restless/irritable, sunken eyes, and a slow skin pinch, which is some dehydration.	Give fluid for some dehydration (Plan B). Advise the mother to continue breastfeeding. Give zinc for 14 days. Follow up in 2 days.	–	
yi_no_dehydration	GREEN – home	There is diarrhoea with not enough signs for some or severe dehydration.	Give fluid and continue breastfeeding at home (Plan A). Give zinc for 14 days. Counsel the caregiver on home care. Follow up in 2 days.	–	

Corrections for Young infant: diarrhoea:

Young infant: congenital problems

Classification	Severity	Trigger (as coded)	Action	Drugs	OK
yi_congenital_priority	PINK – refer	There is a cleft lip or palate, a congenital priority sign.	Give any needed pre-referral treatment. Treat to prevent low blood sugar. Keep the infant warm. Refer URGENTLY to hospital.	–	

Classification	Severity	Trigger (as coded)	Action	Drugs	OK
yi_congenital_abnormal_signs	YELLOW – treat	There is a club foot, an abnormal sign.	Keep the infant warm, skin to skin. Assess breastfeeding and support the mother. Refer for assessment.	–	
yi_possible_congenital_syphilis	YELLOW – treat	The mother's RPR is positive and she was untreated or only partially treated, so congenital syphilis is possible.	Check for signs of congenital syphilis and refer to hospital if present. If there are no signs, give intramuscular penicillin. Ensure the mother receives full treatment.	–	

Corrections for Young infant: congenital problems:

Dosing tables — confirm every band

Doses are not graded by the model scorer; they are emitted verbatim from these tables.

amoxicillin

pneumonia, acute ear infection · oral · two times daily for 5 days · source 2014 p16 · keyed by weight

weight (kg)	tablet_250mg	syrup_250mg_per_5ml_ml	OK
4–10	1	5	
10–14	2	10	
14–19	3	15	

cotrimoxazole__prophylaxis

prophylaxis in HIV confirmed or exposed child · oral · once a day, starting at 4-6 weeks of age · source 2014 p16 · keyed by age

age (months)	syrup_40_200_per_5ml_ml	paed_tablet_20_100	adult_tablet_80_400	OK
0–6	2.5	0.5	0.25	
6–60	5	2	0.5	

paracetamol

high fever (>38.5C) or ear pain · oral · every 6 hours until fever or pain is gone · source 2014 p16 · keyed by weight

weight (kg)	tablet_100mg	tablet_500mg	OK
4–14	1	0.25	
14–19	1.5	0.5	

zinc

diarrhoea (age 2 months up to 5 years) · oral · daily for 14 days · source 2014 (GIVE EXTRA FLUID FOR DIARRHOEA) · keyed by age

age (months)	tablets_daily	OK
2–6	0.5	
6–∞	1	

ors_plan_a_extra_fluid

diarrhoea, no dehydration – amount after each loose stool · oral · · source 2014 Plan A · keyed by age

age (months)	ml_after_each_loose_stool_min	ml_after_each_loose_stool_max	OK
0–24	50	100	
24–∞	100	200	

vitamin_a

supplementation from 6 months; treatment dose for measles or persistent diarrhoea (NOT if dosed in past month or on RUTF) · oral · · source 2014 (Vitamin A supplementation and treatment) · keyed by age

age (months)	iu	OK
6–12	100000	
12–∞	200000	

mebendazole

deworming if hookworm/whipworm endemic, child ≥ 1 year, no dose in previous 6 months · oral · single dose in clinic · source 2014 · keyed by age

age (months)	dose_mg	OK
12–∞	500	

ceftriaxone_im

sick-child pre-referral (very severe disease, suspected meningitis, severe pneumonia, mastoiditis) · IM · · source 2022 p35 · keyed by weight
note: >17.5 kg: give 2 ml in each thigh; for weights over 17.5 kg dilute 1 g in 3.5 ml sterile water and give 5.5 ml IM

weight (kg)	dose_mg	volume_ml	OK
3.5–5.5	312	1.25	
5.5–7	440	1.75	
7–9	625	2.5	
9–11	750	3	
11–14	810	3.25	
14–17.5	1000	4	
17.5–∞	1500	5.5	

diazepam_rectal

convulsing now · rectal · · source 2022 p35 · keyed by weight
note: 0.5 mg/kg per rectum; repeat after 10 minutes if not stopped

weight (kg)	dose_mg	volume_ml	OK
3–4	2	0.4	
4–5	2.5	0.5	
5–15	5	1	
15–25	7.5	1.5	

young_infant_ampicillin_im

young infant (0-2 months) very severe disease – with gentamicin · IM · · source 2014 p51 · keyed by weight
note: 2014 young-infant regimen. Dose 50 mg/kg. 250 mg vial + 1.3 ml sterile water = 250 mg/1.5 ml.

weight (kg)	volume_ml	OK
1–1.5	0.4	
1.5–2	0.5	
2–2.5	0.7	
2.5–3	0.8	
3–3.5	1.0	
3.5–4	1.1	
4–4.5	1.3	

artemether_lumefantrine

uncomplicated malaria (malaria test positive) · oral · two times daily for 3 days · source 2014 p16 · keyed by weight
note: first dose in clinic, observe 1 hour, repeat if vomits within an hour; take with food

weight (kg)	tablets_per_dose	OK
5–10	1	
10–14	1	
14–19	2	

Specific things to check

Confirm every weight/age band boundary against the source table (off-by-one on a band edge is a dosing error).

Ceftriaxone dilution and per-thigh split (>17.5kg) – 2022 p35.

2014 uses ampicillin+gentamicin for the young infant; 2022 sick-child pre-referral is ceftriaxone. Confirm which the national adaptation wants.

ART regimens are edition- and country-specific – verify against current national ART guidelines, not this booklet.

Zinc, ORS volumes, vitamin A, mebendazole are stable across editions but confirm.

Fever severe label kept as very_severe_febrile_disease (2014) with bulging fontanelle added – acceptable?

HIV severity: all tiers MODERATE except hiv_infection_unlikely (MILD), none PINK (ART not urgent) – acceptable?

Young-infant treatment emits no specific dose (IM antibiotics / referral only) – acceptable?

Full per-branch checklist items are in the `src/sft/extended_protocol.py` and `src/sft/young_infant.py` docstrings.

Sign-off

Reviewer name

Signature

Date

National adaptation reviewed against